

1. Q: Why must a full face shield be used when working on the batteries?

A:

- Batteries can explode and emit hydrogen fumes during charging/discharging.
- Hydrogen is extremely explosive — full face shield protects eyes/face.

2. Q: What immediate step must you avoid doing near a charging battery, and why?

A:

- Do not smoke or allow sparks/flames nearby.
- Charging releases explosive hydrogen; prevention avoids ignition.

3. Q: If battery acid contacts skin, what is the recommended first aid?

A:

- Flush externally with water immediately.
- Call a physician if necessary; follow antidote guidance in the manual.

4. Q: How many occupants are MEGA electric vehicles designed to carry?

A:

- Exactly 2: one operator + one passenger.
- Never exceed the rated load capacity on the vehicle ID plate.

5. Q: What are the VIN lengths and where can you find the VIN on the vehicle?

A:

- NEV VIN: 17 digits; LSV VIN: 13 digits.
- Found: glove box, front dash panel, kick panel below driver seat, under steering wheel cover.

6. Q: What does the VIN digit 1–3 (for NEV) represent?

A:

- "5FC" = CPC Manufacturer Identification (per VIN matrix in manual).

7. Q: Give three key vehicle dimensions from the specs table.

A:

- Overall length: 131.1 in.
 - Wheelbase: 91.5 in.
 - Turning radius: 19 ft.
8. Q: What's the vehicle's empty weight and GVWR?

A:

- Empty weight varies by model (examples: 1485 lb, 1474 lb, 1557 lb).
- GVWR (Gross Vehicle Weight Rating): 2464 lb.

9. Q: What is the maximum rated speed and standard battery range?

A:

- Max rated speed: up to 25 MPH.
- Range (standard battery group): up to 40 miles (varies with load/conditions).

10. Q: What motor and traction system specs are listed?

A:

- Traction system: 48 Volt.
- Motor: 5 kW AC with solid state speed control and regenerative braking.

11. Q: What tire size and recommended chassis material appear in the spec sheet?

A:

- Tires: 145/70R13.
- Chassis: Aluminum construction.

12. Q: What does the Multi-Functional Display Indicator (MDI) show when batteries are full or discharged?

A:

- Fully charged: uppermost green LED lit.
- Discharged: lower red LED lit; hour meter shows hours worked or flashes error codes.

13. Q: How do you interpret an MDI alarm code?

A:

- First two digits = electronic module (e.g., 16 = MDI, 02 = traction controller).
- Last two digits = specific alarm code; consult the AC Power System Manual.

14. Q: What must be done before performing any service (power & safety steps)?

A:

- Power key → OFF, directional selector → NEUTRAL, remove key.
- Block tires and turn main disconnect counterclockwise to OFF.

15. Q: When lifting the vehicle, where must the floor jack be placed?

A:

- Under the frame only — never use bumper or cargo bed for lifting.
- Use rated jack stands/support blocks under frame before working underneath.

16. Q: What is the storage tire pressure and storage guidance for vehicles stored 6–8 weeks?

A:

- Maintain tire pressure at 35 psi.
- Store in cool/dry place, clean terminals, continue quarterly lubrication, do not engage parking brake.

17. Q: What's the periodic service calendar example (month → frequency)?

A:

- January: Monthly; March: Quarterly; June: Semi-annual; December: Annual.

18. Q: Describe the maintenance indicator reset procedure in brief.

A:

- Turn keyswitch to position 2 for ~5 s → wrench & mileage show.
- Hold both dash buttons until RESET appears, then cycle keyswitch to confirm.

19. Q: What are the battery acceptance numbers for cell voltage and pack uniformity?

A:

- Each battery should read at least 6.0 V individually.
- No battery should be more than 0.5 V lower than the highest reading in the pack.

20. Q: What capacity/time requirement must a battery pass on load testing?

A:

- Must produce 45 minutes of capacity at a 55 amp draw after full recharge.

21. Q: How often should differential oil be changed initially, and what capacity is required?

A:

- Change oil initially between 25–50 hours.
- Capacity: 34 ounces (1 L); use SAE 30 weight.

22. Q: What is the procedure if brake fluid looks milky or rubber parts are swollen?

A:

- Indicates contamination — drain old fluid, replace cups/seals, flush system, refill with clean fluid.

23. Q: What pedal travel limits indicate brake service is required (periodic inspection)?

A:

- Brake pedal should not travel more than 2½ inches from floor board before firm resistance.
- If pedal is spongy or travels more → service needed.

24. Q: What parking brake handle travel is acceptable before it must be adjusted?

A:

- The handle should ratchet two or more clicks; if it rises > 3½ in, adjust cable/linkage.

25. Q: For rear drum brake service, what minimum lining thickness mandates replacement?

A:

- Replace shoes if lining at any point is < 1/16 in (1.6 mm).

26. Q: When rebore/replace a brake drum — what are the rebore limits and runout warnings?

A:

- Max rebore limit: .060 in over original drum diameter.
- Out-of-round tolerance: do not exceed .006 in total indicator reading.

27. Q: What lug nut torque is specified when reinstalling wheels?

A:

- Tighten lug nuts in crisscross pattern to a maximum of 65 ft·lbs (≈88.1 Nm).

28. Q: What simple checks are on the pre-driving checklist before operating the vehicle?

A:

- Adjust seat & mirror, check battery capacity/MDI, tire inflation, lights/horn/reverse buzzer, and parking brake.

29. Q: How should the charger be connected for storage charging if a Delta-Q charger is present?

A:

- Delta-Q can test/recharge during storage — leave batteries connected and charger plugged to reliable source.

30. Q: What are the two directional knob forward settings and do they differ in speed?

A:

- Move lever to FL or FH to travel forward; both give the same speed.

31. Q: What must be done before charging the batteries with respect to the master key?

A:

- Ensure the master power key switch is OFF and key is removed prior to charging.

32. Q: What are the basic troubleshooting first actions before electrical diagnosis?

A:

- Visually inspect for hazards (sharp edges, open wires, discoloration), record VIN, then fully test batteries.

33. Q: If the MDI displays a flashing error code instead of hours, what should you note?

A:

- Record the error code — it helps identify the module and corrective action per AC Power System manual.

34. Q: How is bleeding the hydraulic brake system performed (order and key caution)?

A:

- Install tubing on bleeder, bleed both right wheels first, then both left; keep master cylinder topped up; close valve when clear of bubbles.

35. Q: When servicing calipers, what's important about piston removal and reassembly?

A:

- Prevent fluid spray (cover caliper), remove pistons using hydraulic pressure, replace seals/boots, clean with alcohol, lubricate with new brake fluid, then bleed system.